

STEMI with Cardiogenic shock



Section I: Scenario Demographics

Scenario Title:	STEMI with Cardiogenic Shock		
Date of Development:	20/06/2015 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Kyla Caners
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To practice managing cardiogenic shock, a relatively rare form of shock.
CRM Objectives:	1) Communicate effectively with team members in the care of a complex, critically ill patient. 2) Prioritize orders, medication administration, and airway management among team members appropriately.
Medical Objectives:	1) Arrange urgent consultant care while managing a critically ill patient. 2) Recognize and appropriately treat cardiogenic shock with inotropes. 3) Demonstrate a safe approach to intubating a patient with cardiogenic shock.

Case Summary: Brief Summary of Case Progression and Major Events

A 55 year old man presents to the ED as a STEMI call. He is profoundly hypotensive with low O2 sats and obvious CHF. The patient's blood pressure will transiently respond to fluid resuscitation. The ECG will show anterolateral ST elevation. The team will need to prepare for intubation while activating the cath lab. They will also need to start vasopressors. The patient will remain hypotensive until an inotrope like dobutamine is initiated. If unsafe medications are chosen for intubation, the patient will have a VT arrest.

References

Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.



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Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Computerized Mannequin	Realism:	☒ Conceptual
	☐ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
<i>Select most important dimension(s)</i>			
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☒ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☒ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
Diaphoresis spray, older man mask.			
E. Approximate Timing			
Set-Up:	3 min	Scenario:	12 min
		Debriefing:	20 min

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Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

A 55-year-old male presents to the ED with EMS as a STEMI activation. He arrives being bagged by EMS for hypoxia. His initial EMS call was for chest pain and he has significantly deteriorated en route. He has a history of smoking, hypertension, diabetes, and hypercholesterolemia. No prior cardiac history.

B. Patient Profile and History

Patient Name: John Hamilton	Age: 55	Weight: 80kg
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Gender: ☒ M ☐ F	Code Status: Full
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Chief Complaint: Chest pain

History of Presenting Illness: Chest pain onset 2 hours prior. Called EMS and they found a STEMI on their field ECG. Patient then started becoming SOB and deteriorated on route.

Past Medical History:	HTN	Medications:	HCTZ 25mg daily
	DM		Metformin 500mg TID
	Dyslipidemia		Lipitor 40mg daily
			Ramipril 5mg daily

Allergies: None.

Social History: Smoker.

Family History: No family history CAD.

Review of Systems:	CNS:	Patient too hypoxic/altered. Unable.		
	HEENT:	Patient too hypoxic/altered. Unable.		
	CVS:	Patient too hypoxic/altered. Unable.		
	RESP:	Patient too hypoxic/altered. Unable.		
	GI:	Patient too hypoxic/altered. Unable.		
	GU:	Patient too hypoxic/altered. Unable.		
	MSK:	Patient too hypoxic/altered.	INT:	Unable.

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display	☒ Monitor On, no data displayed	☐ Monitor on Standard Display
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HR: 110/min	BP: 80/40	RR: 30/min	O ₂ SAT: 85% BVM
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Rhythm: Sinus tach	T: 36.2°C	Glucose: 6.2 mmol/L	GCS: 10 (E3 V2 M5)
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General Status: Cool, clammy, looks unwell.

CNS:	Non-verbal. Depressed LOC.
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HEENT:	Nil.
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CVS:	No murmur.
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RESP:	Diffuse crackles to apices.
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ABDO:	Soft, NT.
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GU:	Nil.
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MSK:	Nil.	SKIN:	Cool, clammy, mottled.
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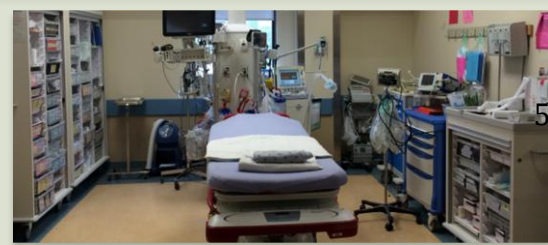
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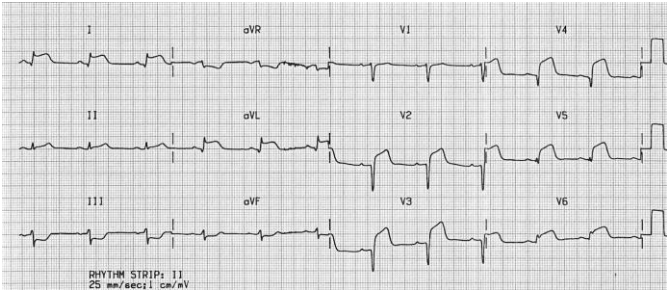
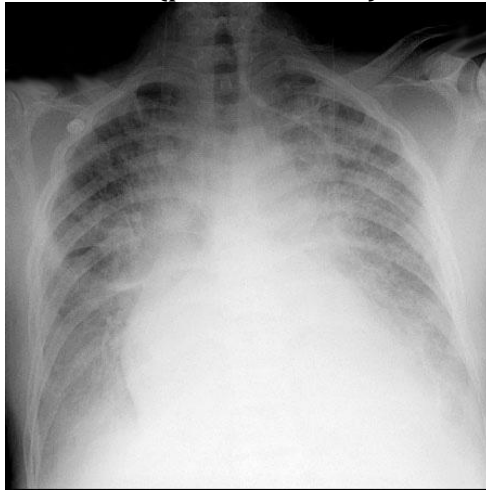
Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus tach HR: 110/min BP: 80/40 RR: 30/min O ₂ SAT: 85% BVM T: 36.2°C	Looks moribund. Cool, clammy, mottled skin. GCS 10 (E3 V2 M5).	<u>Learner Actions</u> ☐ IV access, O ₂ , monitors ☐ ECG (STEMI) ☐ Blood work ☐ Trial 250-500ml NS bolus ☐ Check cap sugar (6.2) ☐ Prepare for intubation ☐ Call cath lab (they will be delayed 30 min)	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Bolus given → BP 85/45 - 5 minutes → desat to 80% <u>Triggers</u> <i>For progression to next state</i> - Intubation → 2. Intubation
2. Intubation As above.	As above.	<u>Learner Actions</u> ☐ Consider bipap as bridge ☐ Pre-medicate with fentanyl ☐ Induction with etomidate or ketamine ☐ Use of paralytic ☐ Vasopressor at bedside ☐ PEEP valve ready	<u>Modifiers</u> <u>Triggers</u> - Propofol used or unsafe intubation → 3. VT arrest - Intubation safe → 4. Ongoing hypotension
3. VT arrest Rhythm → VT HR → 180 BP → ?/?	Patient pulseless and unresponsive.	<u>Learner Actions</u> ☐ High quality CPR. ☐ Defibrillate VT. ☐ Amiodarone 300mg iv ☐ Epinephrine 1mg iv q3min	<u>Modifiers</u> <u>Triggers</u> - 2 cycles CPR → 4. Ongoing hypotension
4. Ongoing hypotension HR → 120 BP → 70/30 O ₂ SAT → 92% on 100%	Intubated and sedated.	<u>Learner Actions</u> ☐ CXR (post-intubation) ☐ Sedation ☐ OG insertion to facilitate giving ASA and ticagrelor ☐ Heparin 4000 units iv ☐ Call cath lab again ☐ Start norepi infusion at 0.05-2 mcg/kg/min and titrate up ☐ Start dobutamine infusion at 2.5mcg/kg/min ☐ Consider another 500ml NS	<u>Modifiers</u> - Dobutamine started 1st → BP 50/10 - Norepi started → BP 75/35 - Norepi titrated up 2-3 times → BP 85/45 - Dobutamine added → BP 90/55 <u>Triggers</u> - 12 min → 5. Stabilization - Vasopressor & inotrope started → 5. Stabilization
5. Stabilization HR → 100 BP → 95/65 O ₂ SAT → 92% on 100%	As above.	<u>Learner Actions</u> ☐ Finish tasks from above	Cath lab calls to say they are ready for the patient. END CASE

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results	
No lab results required for this case.	
Images (ECGs, CXRs, etc.)	
ECG – anterolateral STEMI  ECG source: https://lifeinthefastlane.com/ecg-library/lateral-stemi/	CXR – CHF (pre-intubation)  CXR source: https://www.med-ed.virginia.edu/courses/rad/cxr/pathology2Bchest.html
CXR – CHF (post-intubation)  CXR source: https://heart-conditions.knoji.com/learning-about-and-coping-with-congestive-heart-failure/	
Ultrasound Video Files (if applicable)	
Echo showing RWMA	Lung U/S – pulmonary edema

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) What did you see as the priorities for this patient's management on presentation? How did you juggle these and decide which came first? 2) Describe the team dynamic and communication in the room. Do you feel the patient got everything he needed? Was the environment calm? 3) Do you have an approach to treating cardiogenic shock? 4) Do you have an approach to intubating a patient with cardiogenic shock? What are some of the risks/benefits of the different agents available for induction? What else is important to consider as your prepare for intubation in this patient? What equipment should you have available?	
Key Moments	
Intubation.	
Activation of cath lab.	
Treatment of cardiogenic shock.	